

Clinical, Laboratory, and Molecular Characteristics of Inherited Vitamin K–Dependent Coagulation Factors Deficiency

Salvatore Perrone, MD^{1,*} Simona Raso, MD, PhD^{2,*} Mariasanta Napolitano, MD, PhD³ 

¹Department of Hematology, S. M. Goretti Hospital, Polo Universitario Pontino, Latina, Italy

²Department of Hematology and Rare Diseases, V Cervello Hospital, Azienda Ospedaliera Ospedali Riuniti Villa Sofia-Cervello, Palermo, Italy

³Department of Health Promotion, Mother and Child Care, Internal Medicine and Medical Specialties (PROMISE), University of Palermo, Palermo, Italy

Address for correspondence Mariasanta Napolitano, MD, PhD, Department of Health Promotion, Mother and Child Care, Internal Medicine and Medical Specialty (PROMISE), University of Palermo, Via del Vespro 129, 90127 Palermo, Italy (e-mail: mariasanta.napolitano@unipa.it).

Semin Thromb Hemost 2025;51:170–179.

Abstract

Vitamin K–dependent coagulation factors deficiency (VKCFD) is a rare autosomal recessive genetic disease characterized by impaired levels of multiple coagulation factors (II, VII, IX, and X) and natural anticoagulants (proteins C and S). VKCFD is part of familial multiple coagulation factor deficiencies, reporting overall 50 affected families thus far. Disease manifestations are quite heterogeneous, bleeding symptoms may vary, and even, although generally mild, some patients may succumb to fatal outcomes. VKCFD diagnosis may be delayed because the disease phenotype simulates the most frequently acquired deficiencies of vitamin K. First-line coagulation assays, prothrombin time/international normalized ratio (PT/INR) and activated partial thromboplastin time (aPTT), are both prolonged; mixing test typically normalizes the clotting times; and vitamin K–dependent coagulation factors will be variably decreased. Molecularly, VKCFD is associated with mutations in *γ-glutamyl-carboxylase* (*GGCX*) or *vitamin K epoxide reductase* complex subunit 1 (*VKORC1*) genes. Vitamin K is involved not only in the biosynthesis of coagulation proteins but also in bone metabolism and cell proliferation. Therapeutic options are based on vitamin K supplementation, coagulation factors (prothrombin complex), and fresh frozen plasma, in case of severe bleeding episodes. Two case studies here illustrate the diagnostic challenges of VKCFD: case 1 depicts a woman with a history of bleeding episodes, diagnosed, only in her third decade of life with inherited homozygous *GGCX* gene mutation. Case 2 shows a man with an acquired vitamin K deficiency caused by Crohn's disease. Better understanding of *GGCX* and *VKORC1* mutations aids in prognosis and treatment planning, with emerging insights suggesting potential limitations in the effectiveness of vitamin K supplementation in certain mutations.

Keywords

- VKCFD
- γ -Glutamyl carboxylase
- VKORC1
- rare bleeding disorders
- familial multiple factor deficiencies
- vitamin K antagonists

* These authors contributed equally to this article.

article published online
November 4, 2024

Issue Theme Precision Medicine in Rare Bleeding Disorders; Guest Editors: Maha Othman, MD, PhD, and Akbar Dorgalaleh, PhD

© 2024. Thieme. All rights reserved. Thieme Medical Publishers, Inc., 333 Seventh Avenue, 18th Floor, New York, NY 10001, USA

DOI <https://doi.org/10.1055/s-0044-1792031>.
ISSN 0094-6176.

Vitamin K–dependent coagulation factors deficiency (VKCFD) is a very rare autosomal recessive genetic disease characterized by multiple decrease in the levels of several coagulation factors, involving the glutamination, vitamin K (VK) mediated, of coagulation factors II (FII), VII (FVII), IX (FIX), and X (FX), as well as natural anticoagulants protein C (PC), protein S (PS), and protein Z (PZ). Two separate genes are involved in the pathogenesis of this disease: γ -glutamyl-carboxylase (*GGCX*, OMIM 277450) and vitamin K epoxide reductase complex subunit 1 (*VKORC1*, OMIM 607473).^{1–3} Patients harboring defects in *GGCX* have VKCFD type 1, while those with *VKORC1* defects have VKCFD type 2. For classification purposes, VKCFD is part of the familial multiple coagulation factor deficiencies (FMCFDs), a group of rare inherited disorders characterized by a simultaneous decrease in the levels of two or more coagulation factors arising from a genetic defect or defects.⁴ The prevalence of VKCFD is unknown, but less than 50 affected families and 272 individuals with publicly available variants have been reported worldwide in the literature to date.^{5,6} Indeed, all available information are mainly based on multiple case reports and small clinical series,^{2,3,7–37} the first identified in 1966.³² Clinical manifestations of VKCFD are variable from one individual to another; they typically involve not only the coagulation system but also connective tissue's health and cell signaling. Indeed, VK is involved in the carboxylation of several proteins in different tissues and organs: K2-dependent proteins activate a protective mechanism preventing the development of vascular calcification involved in the occurrence of cardiovascular disease³⁸ and patients treated with VK inhibitors develop calcifications of vessels and cardiac valves.³⁹ It is thus hypothesized that lectin and adiponectin are involved in an intricate network for glucose metabolism and osteocalcin promotes proliferation of pancreatic β cells; so, the implications of these proteins in diabetes are under active research.^{40,41} VK is also implicated in inhibiting the proliferation, growth, and differentiation of cancer cells through various mechanisms, including induction of *c-myc* and *c-fos* genes, regulation of B cell lymphoma-2 (*Bcl-2*) and *p21* genes, and angiogenesis inhibition⁴²; therefore, its involvement in cancer has been suggested.⁴³ With reference to the hemostatic profile of VKCFD, bleeding symptoms are generally mild. In the most severe forms of the disease, symptoms appear at birth, while in milder forms their onset may be later in life. Bleeding manifestations at birth must be differentiated from VK deficiency of the newborn,⁴⁴ while symptoms with late onset in adulthood are more frequently related to acquired forms secondary to disorders of the bowel,⁴⁵ liver disease,⁴⁶ dietary VK deficiency,^{47,48} or drugs^{49,50} like VK antagonists.⁵¹ Subjects affected by VKCFD may show other nonbleeding abnormalities due to defective carboxylation in extracellular connective tissue proteins such as osteocalcin and Gla proteins, including midfacial hypoplasia, premature osteoporosis, cochlear hearing loss, heart valve defects, pulmonary stenosis, or pseudoxanthoma elasticum-like phenotype.⁵²

The primary aim of this narrative review is to analyze, starting from two paradigmatic cases, the scientific data on

the clinical, laboratory, and molecular aspects characterizing VKCFD. We highlight current and future challenges in the diagnostic workup and treatment choices of VKCFD.

Case 1

A 34-year-old Caucasian woman was referred to our hemophilia center, for prolonged clotting times incidentally found before an elective colonoscopy: the activated partial thromboplastin time (aPTT) was mildly prolonged (37 seconds) and the prothrombin time/international normalized ratio (PT/INR) was remarkably increased (INR = 3.63); fibrinogen, D-dimer, and platelet counts were normal. She had a previous history of repeated nose and gum bleedings, excessive menstrual blood loss, severe postpartum hemorrhage requiring blood transfusions, and hemoperitoneum cyst rupture. She had also received a diagnosis of precocious “osteoporosis.” The family history was negative for hemorrhagic diathesis. Despite a personal history characterized by several, even severe, bleeding symptoms with a bleeding score of 6, calculated by International Society on Thrombosis and Haemostasis Bleeding Assessment Tool (ISTH-BAT),⁵³ she was never investigated for hemorrhagic disorders. A revision of her past exams showed the presence of abnormal standard coagulation assay results over time, although with variable values, and an anecdotal positive lupus anticoagulant (LA) assay.

At first step, PT/INR and aPTT assays were repeated in the hospital clinical laboratory (HCL), confirming the abnormal results. The second laboratory step was taken, according to clinical suspicious and past bleeding history, by performing a mixing study, and, following its correction, VK-dependent factors with the following results: FVII:C = 10% (activity), FIX:C = 21%, FX:C = 8%, FII:C = 13%, PC = 34% (normal range: 70–140%), and PS = 21% (normal range: 74–146%).

Protein C was measured on an ACL 9000 Coagulometer (Instrumentation Laboratories, Lexington, MD) using IL Test ProClot (Instrumentation Laboratory) reagent. Protein S was measured on an ACL 3000 Coagulometer by the method of IL-Test Protein S (Instrumentation Laboratory).

In the presence of a laboratory diagnosis of VKCFD, the genetic defect was then investigated by Sanger sequencing, and a pathogenetic homozygous mutation in the γ -glutamyl-carboxylase (*GGCX*) gene was found. To prepare the patient for an invasive procedure, an intravenous (IV) single dose of vitamin K1 (10 mg) was administered 1 hour before the colonoscopy; this allowed a fast, efficient, and sustained normalization of the both coagulation times (PT and aPTT).⁵⁴ The patient did not develop any procedure-related complication and she was started on prophylactic treatment with twice weekly oral VK with the purpose of reducing her excessive menstrual blood losses.

Case 2

A 45-year-old Caucasian man presented to the emergency room of our hospital, due to a recent onset of extensive bruising and widespread subcutaneous hematomas; he had no family or personal history of previous bleeding episodes,

even after a hernia surgery procedure. He was on oral medication (ACE inhibitors) for high blood pressure, and he did not take any additional drug recently. Moreover, the patient complained of diarrhea and rare episodes of rectal bleeding in the last 2 months. The first-line coagulation assays showed a nondetectable PT/INR and a considerably prolonged aPTT (57 seconds) while fibrinogen, D-dimer levels, and platelet count were normal. He underwent second-level coagulation assays in HCL, including a mixing test, LA assay, platelet function tests, levels of coagulation factors, and fibrinolytic proteins (plasminogen, α 2-antiplasmin). Second-level coagulation assays revealed decreased VK-dependent factors levels: FII:C=17%, FVII:C=2.9%, FIX:C=11%, FX: C=8.5%, PC=58%, PS=64%; AT=94%, and plasminogen 116% (normal range: 80–130%), α 2-antiplasmin=108% (normal range: 98–122%).

Platelet function testing (PFA100) was normal and LA was negative. Based on laboratory findings and patient history of recent onset of bleeding symptoms, an acquired deficiency of VK factors was suspected. To define the etiology of acquired VK deficiency, further laboratory assays were performed including liver function tests and fecal calprotectin. Results of these investigations showed an increase in C-reactive protein (22 mg/dL), gamma-globulin plasma concentration (3.45 g/dL), and a high level of fecal calprotectin (504 μ g/g). Endoscopic examinations of the gastroenteric tract (gastroscopy and colonoscopy with biopsy) led to the diagnosis of Crohn's disease causing VK deficiency due to malabsorption.⁵⁵

During the acute phase of Crohn's disease, intravenous VK1 was regularly administered at the dosage of 10 mg to control rectal bleeding. Parenteral VK treatment induced a fast, efficient, and sustained normalization of coagulation times and controlled clinical symptoms; after the acute phase, VK treatment was discontinued.

The Hemostatic and Nonhemostatic Role of Vitamin K

VK is a lipo-soluble vitamin involved in the process of carboxylation of several proteins including coagulation factors; among VK-dependent nonhemostatic proteins, there are matrix Gla protein (MGP), osteocalcin (BGLAP), proline-rich Gla proteins (PRGPs) 1 and 2, upper zone of the growth plate and cartilage matrix-associated protein (UCMA/GRP), transmembrane Gla proteins (TMGs) 3 and 4, and growth arrest-specific 6 (GAS6 matrix-Gla protein). These proteins have several functions: BGLAP, MGP, and GRP regulate the calcification processes of bone and soft tissues. GAS6 is involved in cell signaling and platelet aggregation.^{30,34,36}

VK1 (phylloquinone) is present in fruits and vegetables (in particular green leafy vegetables, herbs, vegetable oils), and represents the main source of VK in diet (90%). VK2 (menaquinones) is found in animal-based and fermented foods or produced by intestinal bacteria, but its total amount is quite limited.⁵⁶ In Europe, the recommended VK daily allowance is 75 μ g (Commission Directive 2008/100/EC).⁵⁷ Additionally, VK2 has been suggested to be suitable for a specific dietary recommendation intake.⁵⁸ Vitamin K2 has a higher bioactivity than vitamin K1 for several molecular processes, including γ -glutamylcarbox-

ylation activity, inhibitory effect on bone resorption, antioxidant, and anticancer effects.^{43,59}

The process of carboxylation consists in the posttranslational modification of some particular glutamate (Glu) amino acids that receive a carboxylic group to form a carboxyl glutamate (Gla), which generates a calcium-binding module that is essential for the activity of the protein.⁶⁰

Glu is targeted for carboxylation because they contain an exosite binding domain (EBD) that can be recognized by the enzyme. The GGCX, which is an integral membrane protein localized into the endoplasmic reticulum, allows interactions with the hydrophobic VK and generates Gla residues by the oxygenation of reduced VK, which in turn becomes converted to VK epoxide during this reaction (**Fig. 1**).⁶¹ In particular, GGCX catalyzes a reaction between oxygen and reduced VK to generate a VK superbase that extracts a hydrogen from Glu, thus generating a Glu carbanion that then reacts with CO₂ to form Gla.⁶² The VK epoxide produced by GGCX must be reduced by VKORC to regenerate VK hydroquinone, which is required for continual carboxylase activity.⁶³ VKORC is also targeted by warfarin; therefore, when VK hydroquinone levels are decreased, carboxylase activity is disrupted and, in turn, levels of active VK-dependent proteins including those involved in hemostasis are decreased; this explains the clinical activity of VK antagonist drugs.⁶⁴ Hereditary mutations in *VKORC1* cause warfarin resistance and multiple coagulation factor deficiency type 2.⁶⁵ Indeed, in clinical practice, VK is used to reverse VK antagonist overdose or rodenticide poisonings,⁶⁶ and it is also administered in newborns to prevent the classic hemorrhagic disease of the newborn.⁶⁷

Molecular Basis of VKCFD

GGCX is a well preserved and not polymorphic gene; indeed, only two variants (rs699664 and rs6173310) of the 400 known variants have an allelic frequency >0.0001.⁶⁵ Mice lacking the homologous GGCX protein show a complete lethality secondary to hemorrhages, suggesting the critical role of GGCX and the lack of alternative pathways.⁶⁸ Only 40 gene mutations occurring in GGCX have been detected by genetic screenings of patients with VK-related disorders (**Fig. 2**). De Vilder et al revised available cases and tried to correlate the genomic and phenotypic spectrum of mutations in GGCX gene. Beyond coagulation, other systems can be involved: bones, cardiac, skin, and ocular systems. In particular, VKCFD1 can be associated with skeletal (midfacial hypoplasia, reduced bone mass, chondrodysplasia punctata) or cardiac abnormalities (patent ductus arteriosus Botalli, septal closure defects) in some patients.^{69,70} In the systematic review from De Vilder, 47 patients with mutations of GGCX were identified; of these, 33 patients had VKCFD1, 25 of which had a deficiency of all VK-dependent coagulation factors. Reduced FX activity was observed in all patients, FII impairment in 30 patients, FVII deficiency in 31 patients, and FIX was abnormally low in 24 patients. Of note, FIX deficiency was present only in those patients in whom all the other coagulation factors were also deficient. Of the 33 patients with VKCFD1, 21 patients were symptomatic (spontaneous mucosal bleeding, intra-articular bleeding, abnormal bleeding after injuries, vaccination, and surgery). Ten patients were severely affected with

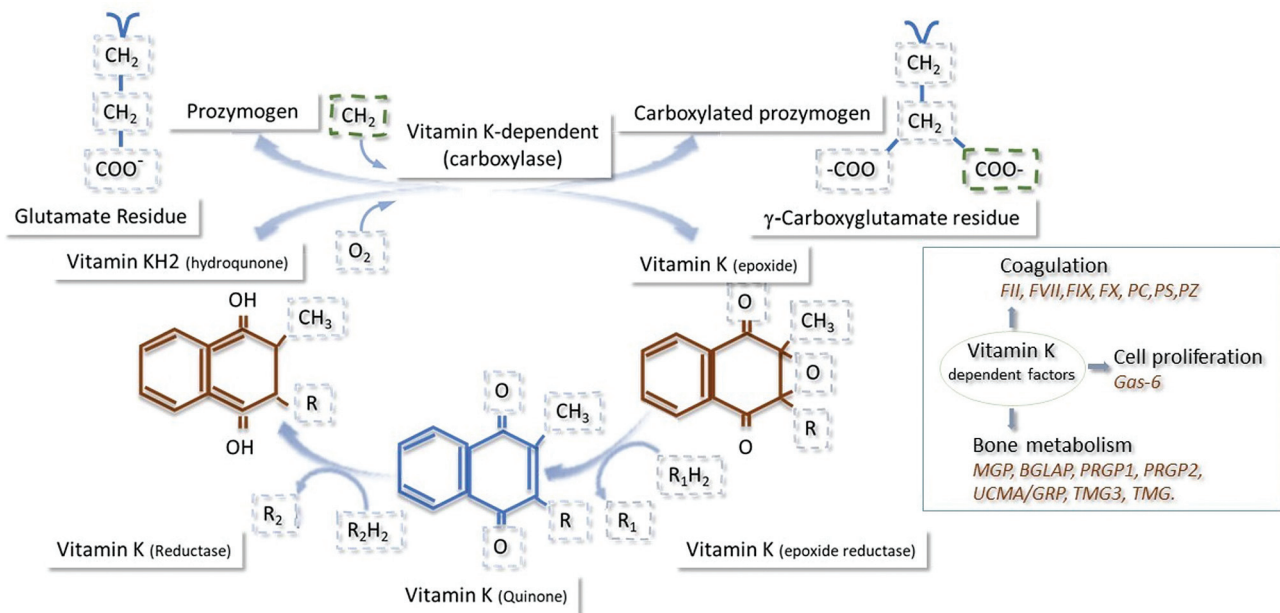


Fig. 1 Vitamin K metabolism and function in coagulation and beyond. Coagulation: factor II (FII), factor VII (FVII), factor IX (FIX), factor X (FX), protein C (PC), protein S (PS), protein Z (PZ). Beyond coagulation: matrix Gla protein (MGP), osteocalcin (BGLAP), proline-rich Gla proteins (PRGPs) 1 and 2, upper zone of the growth plate and cartilage matrix-associated protein (UCMA/GRP), transmembrane Gla proteins (TMGs) 3 and 4, and growth arrest-specific 6 (GAS6).

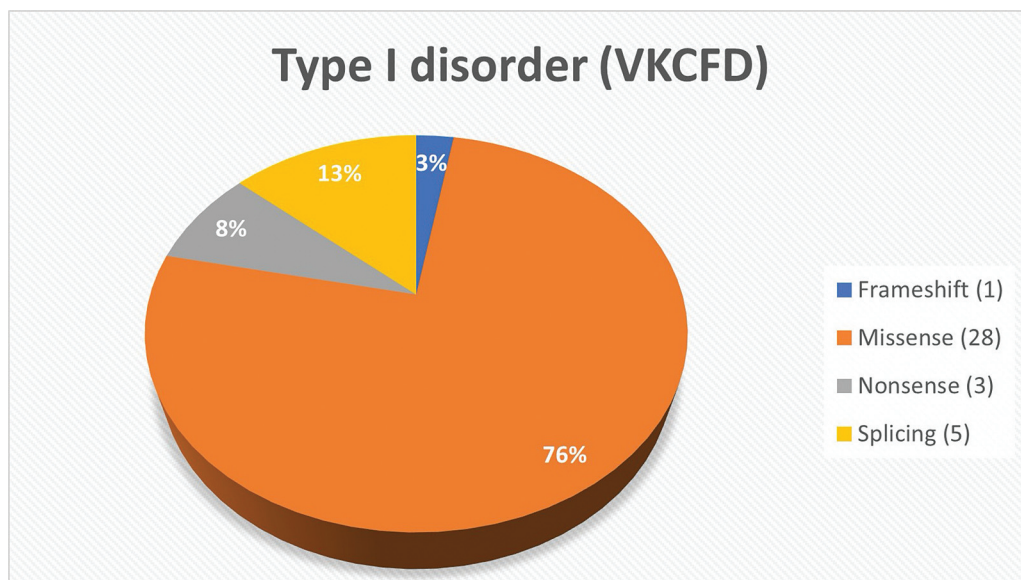


Fig. 2 This diagram shows the relative frequency of reported cases of mutations involving GGCX. To date, at least 34 mutations have been reported in the GGCX gene, which are associated with VKCFD and the majority of them are point mutations. The mutations can be observed in homozygous or compound heterozygous.

symptoms onset before the age of 1 year: all had a combined deficiency of FII, FVII, FIX, and FX; none of the severely affected patients had cutis laxa or other pseudoxanthoma elasticum (PXE)-like⁷¹ skin or eye manifestations. Eleven patients developed a bleeding, four of them had skin symptoms and three had eye manifestations.⁶⁹

In two unrelated patients with VKCFD type 2, the same homozygous point mutation was identified by Rost et al in the third exon of the *VKORC1* gene (located on chromosome 16p11.2): a 292C-T transition resulting in an arg98-to-trp (R98W) substitution.⁶⁵ The families of the two subjects were

of Lebanese and German origin.¹¹ Their haplotypes in the region of homozygosity encompassing the mutated gene were different. Furthermore, cytosine-292 is part of a CpG dinucleotide, which is a known mutation hotspot. Rost et al suggested that gene mutation in these families probably arose independently.⁶⁵ Subsequently, Marchetti et al reported a patient with moderate to severe bleeding tendency and found to be homozygous for the unique *VKORC1* mutation (Arg98Trp) so far detected in VKCFD2. This R98W mutation causes mislocalization from endoplasmic reticulum and degradation of the VKORC protein, and thus reduces most of its activity.⁵⁴ This patient also carried a homozygous

FVII gene polymorphism (613878.0013) associated with further reduction in factor VII levels.⁷²

Clinical Manifestations of VKCFD

Even if a reduced activity of natural anticoagulants might be potentially associated with thrombotic risk, there are no reports of venous or arterial thrombosis in patients affected by VKCFD, commonly presenting, on the contrary, with bleeding symptoms and a quite variegate clinical phenotype.⁷³

Most patients seek medical attention for bleeding episodes ranging from mild bruises,²³ or epistaxis²⁴ to severe intracranial bleeding.³⁴ The vast majority of available reports are from pediatric patients and focused on prenatal diagnosis, familial screening, and genetic counseling^{11,36,74}; however, cases of prolonged bleeding after surgical procedures are also described.³³

This last clinical scenario was described in case 1 with major bleeding episodes occurring mainly postpartum and after the rupture of ovarian cysts. However, the patient also experienced spontaneous, mild hemorrhagic symptoms.

The case underlines how VKCFD is defined by a wide spectrum of symptoms. Usually, the cluster of symptoms involves skin and mucosae with frequent easy bruising. Symptoms occur either spontaneously or in a surgical or trauma setting. Mucocutaneous bleeding, such as gastrointestinal bleeding, may also appear in conjunction with antibiotic therapy, as a consequence of the reduced VK2 production by gut bacteria.^{1,42}

VKCFD can sometimes cause fatal intracranial bleeding in the first weeks of life, thus resembling the acquired hemorrhagic disease of the newborns.^{10,32,34} Umbilical cord bleeding has also been described,^{32,73} while hemarthrosis seems quite rare.⁷³

Clinical burden depends on the residual coagulation factor activity, which is also based on the availability of VK.¹⁶ Moreover, antibiotic and anticonvulsant therapy administration must be carefully evaluated, as these drugs can contribute to worsen the bleeding pattern.⁷³

VKCFD-affected patients may also suffer from non-hemostasis-related symptoms, including mental retardation, osteoporosis, increased fracture rates, and risk of cardiovascular diseases, due to the defective γ -carboxylation of several other VKD proteins.

In case 1, the pattern of VKCFD symptoms included “precious osteoporosis” in the absence of any additional risk factor.

Skeletal abnormalities of subjects affected by VKCFD include nasal hypoplasia, distal digital hypoplasia, and epiphyseal stippling; they all resemble those observed in warfarin embryopathy.⁸ Pseudoxanthoma elasticum-like (PXE-like) disorders have also been reported in patients carrying *GGCX* gene mutations⁷¹ leading to skin hyperlaxity and folds characterized by severe fragmentation and calcification of elastic fibers. Cardiac abnormalities including calcified peripheral arteries, subclinical atherosclerosis, and congenital atrial septal defects are also detected in PXE-like disease. An increased rate of fetal loss has been hypothesized by some authors as a result of bone metabo-

lism during embryogenesis; however, the exact incidence of abortion is unknown due to the rarity of the disease.^{2,8}

Laboratory and Differential Diagnosis

Case 1 highlights that VKCFD is a very rare hemorrhagic disorder for which the clinical suspicion is crucial; the diagnostic delay observed may be due to several reasons. First, it is possible that the hemorrhagic phenotype of the woman was considered mild and mainly due to concomitant risk factors; another potential reason for this delay may be due to the variable prolonged values of first-line coagulation assay results (PT/INR and aPTT), probably attributed to the detection of LA.

While the diagnosis of a single-coagulation factor deficiency is often straightforward, the diagnosis of VKCFD can be challenging due to a wide heterogeneity of presentations and often indistinct clinical and laboratory features.

Facing a clinical suspicion of coagulation disorder, a fundamental step is to collect an accurate personal bleeding history and to systematically evaluate the severity of bleeding symptoms by adopting BATs.⁵³

The patient from case 1 had an abnormal BAT score highlighting the need for further investigations. Familial history may be negative because the defect is autosomal recessive even if carriers of genetic missense mutations in *GGCX* or *VKORC*⁷⁵ may be symptomatic. The diagnostic workup requires the exclusion of other, more frequent, conditions mimicking VKCFD. Initial laboratory finding is characterized by prolonged PT/INR and aPTT. The differential diagnosis of a prolonged coagulation time assay includes artifacts such as a high hematocrit, deficiency of one or more clotting factors, the presence of inhibitors, therapeutic anticoagulation, or sample contamination with anticoagulant medications. The most common cause of a prolonged aPTT reported in the acute care setting is LA, while factor deficiencies are rare.⁷⁶

Liver failure, disseminated intravascular coagulation, or defects in the common coagulation pathway are warranted in the differential diagnosis.⁷⁷

In VFCKD, the extent to which the PT and aPTT are prolonged may be influenced by the degree of individual coagulation factor reduction.^{75,78} Usually, PT/INR is slightly more prolonged than aPTT because FVII has the shortest half-life. VK-dependent coagulation factors activity can be variably decreased, usually ranging between 20 and 60% and less commonly below 10%.⁷⁹

Abnormal screening coagulation tests are followed by mixing studies (50:50) to rule out an inhibitor's interference like in acquired hemophilia or in the very rare presence of inhibitors against FVII (**► Fig. 3**). When mixing studies correctly, as in case of VKCFD, specific factor assays are performed to identify the deficiency.⁸⁰

Other rare inherited bleeding disorders may mimic the clinical presentation of VFCKD and they should be kept in mind in differential diagnosis. These include, in order of frequency, single coagulation defects of FIX (hemophilia B); FVII deficiency; the rarest combined FII and FX deficiencies; as well as other combined defects like combined FVII and FX deficiency or type 2N von Willebrand disease or a FV disorder with overexpression of FV short.^{78,81}

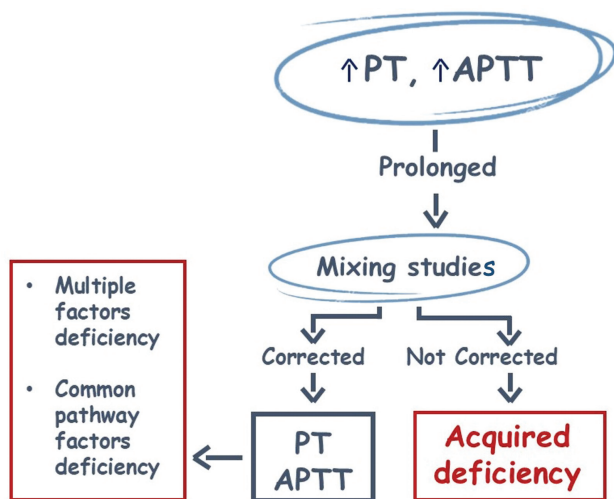


Fig. 3 Laboratory approach to the differential diagnosis of VKCFD. Prolongation of PT and aPTT requires a mixing study with normal plasma, a correction of both assays after mixing implies that multiple factor deficiencies are present while persistently prolonged PT and aPTT suggest inhibitors or acquired conditions like high titer lupus anticoagulant or severe disseminated intravascular coagulation.

Coagulation assays are extensively adopted in clinical practice as indicators of vitamin K deficiency; however, this approach lacks sensitivity and specificity because it mirrors only markedly decreased VK-dependent coagulation factors. Usually, an approximate 50% decrease in prothrombin levels (FII) is required for prolonging PT values.⁸² The measurement of phyloquinone (K1) concentrations in serum is a marker of vitamin K status and it reflects abundance of the vitamin. Concentrations of K1 below 0.15 µg/L are suggestive of deficiency, but they do not necessarily mirror its use by target tissues.⁸³ Moreover, the disadvantages of this approach include exclusion of the other vitamin K homologues, interference by dietary lipid content, and a not-easy-to-access to the assay.⁸⁴

Oxidized vitamin K (KO) is typically undetectable in VKCFD type I, even after vitamin K supplementation, but in VKCFD type II, an increase in KO serum levels can be observed following VK supplementation. In increased protein induced by VK absence/antagonism (PIVKA-II), also called Des-gamma-carboxy prothrombin (DCP), circulating levels can be detected in case of vitamin K deficiency. PIVKA-II or DCP levels can thus be adopted as an indirect indicator of hepatic vitamin K status, and their abnormal levels precede any subsequent change in PT.⁸⁵ PIVKA-II can be measured by automated immunoassay, manual ELISA, or liquid chromatography-tandem mass spectrometry (LC-MS/MS) and show minimal variability for age or dietary lipid content.^{85,86} In patients with hemorrhagic phenotype, VKCFD diagnosis is based on disproportional bleeding symptoms in comparison to the reduced levels of the affected clotting factors. A clinical distinction is, however, not always easy and requires a precise laboratory assessment as well as genotype confirmation. As shown in case 2, it is essential to differentiate between acquired VK deficiencies and VKCFD. The acquired forms can be attributable to (1) intestinal malabsorption of

VK (inflammatory bowel diseases or celiac disease); (2) severe liver dysfunction (e.g., liver cirrhosis); (3) drug interactions caused by voluntary or accidental ingestion of warfarins and superwarfarins; (4) hemorrhagic disease of the newborn.⁷⁹ Case 2 showed an atypical early acquired form of VK deficiency due to Crohn's disease onset; VK deficiency in fact usually occurs late during the clinical course of Crohn's disease and is associated with abnormal bone metabolism.⁵⁵ Indeed, deficit of dietary VK can be rarely observed,⁸⁷ especially in newborns whose parents refuse prophylactic administration of VK at birth.⁸⁸ Interference in metabolism of VK and antibiotics has also been reported.⁸⁹ In addition, surreptitious use of VK antagonist and suicidal use of rodenticides⁹⁰ must be accurately evaluated; the diagnostic work-up should include a thorough psychosocial and psychiatric assessment, to rule out attempted suicide and factitious purpura.⁶⁶ Unfortunately, toxicology testing with liquid chromatography–mass spectrometry (LC-MS/MS) may be unavailable in clinical laboratories of community hospitals.⁹¹ Re-testing after administration of oral or parenteral doses of VK can show transient shortening of PT/INR and aPTT, with consequent increased expression of VK-dependent coagulation factors. Typically, in VKCFD, coagulation parameters improve after VK administration, but after a few hours or days this beneficial effect is lost.⁴

When VKCFD is highly suspected, a research laboratory can perform genotyping of *GGCX* and *VKORC1* genes to confirm the diagnosis.⁹²

The genotyping of the two involved genes *VKORC* (spanning 5 kb and containing 3 exons on chromosome 16)⁹³ and *GGCX* (spanning 13 kb and containing 15 exons on chromosome 2)⁹⁴ can be performed in selected laboratories, to confirm the homozygous presence of mutations. Gene analysis is mandatory to exactly define a complex coagulation defect that must be distinguished from many acquired conditions as well as from other single or multiple congenital coagulation defects. In selected cases with severe bleeding, the study of the pedigree of the family, encompassing genetic and psychological counseling, is advised. On a molecular basis, we can distinguish two forms of the disease: VKCFD type 1 (VKCFD1) presenting mutations in *GGCX* and VKCFD type 2 (VKCFD2) presenting mutations in *VKORC* gene.^{95,96} To date, prenatal diagnosis of VKCFD is not suggested, considering the potential hemorrhagic risk of the procedure and the prevention of major hemorrhagic events in the affected newborn by the routine replacement therapy with VK1 in the third trimester of pregnancy in mothers which are at risk of VKCFD. However, the antenatal management and timing of replacement therapy to prevent intrauterine bleeding and bone or skin manifestations of the disease should be implemented.³⁰

Therapeutic Considerations

The first explored and approved therapeutic option adopted since the discovery of VKCFD consists in the supplementation with exogenous phyloquinone (VK1), since clotting times improve after its administration (→Table 1). However, the exact mechanisms by which VK bypasses the gene defect are

Table 1 Treatment of vitamin K–dependent coagulation factors deficiency

Treatment	Dosage	Schedule	Route of administration	Indication
Vitamin K1 ⁷⁹	5–20 mg	According to PT/INR	IV or oral	Minor bleeding
Tranexamic acid ^{79,97}	15–20 mL/kg or 1 g	Up to every six hours	IV or oral	Minor bleeding
FFP ⁹⁸	15–20 mL/kg	According to clinical situation	IV	Severe bleeding
3 or 4 PCCs ⁹⁹	20–30 units/kg	According to clinical situation	IV	Severe bleeding and major surgery
rFVIIa ^{33,103}	10–20 µg/Kg	According to clinical situation	IV	Severe bleeding and major surgery

Abbreviations: FFP, fresh frozen plasma; IV, intravenous; PCCs, prothrombin complex concentrates; rFVIIa, recombinant activated FVII.
 Note: Data on recommended trough levels for each treatment are not available.

not fully understood, and it is not clear how and if the nonbleeding manifestations of the disease can be affected by prolonged exposure to VK (i.e., by impairing the vascular calcification process).⁷⁹

Oral administration of VK1 may partially correct low coagulation factor levels of approximately 15 to 20%, but it may not be able to prevent major bleeding.^{1,79} In cases where oral treatment is not well tolerated or ineffective, intravenous infusion of VK1 can be considered as a valid option, and the schedule of treatment could vary according to PT/INR values. However, a specific VK treatment schedule has not been yet approved for VKCFD. In milder forms or in minor surgery setting, tranexamic acid can be administered as single agent at the dosage of 15 to 20 mL/kg or 1 g every 6 hours.^{79,97} During major surgical procedures and severe hemorrhage, fresh frozen plasma (FFP) infusion is indicated; usually, multiple infusions are needed at the recommended dose of 15 to 20 mL/kg of body weight.⁹⁸

As an alternative to FFP, three or four prothrombin complex concentrates (PCCs) should be considered,^{78,79,97} although data on use in VKCFD are extremely limited.⁹⁸ PCCs have been effectively adopted in warfarin reversal; thus, available schedules derive from this indication, with an initial dose of 500 units (U) given intravenously for a PT/INR below 5.⁹⁹

With reference to the potential risk of transmission of blood-borne pathogens, even if theoretically possible, it is extremely low¹⁰⁰ due to viral inactivating procedures, while a low but appreciable thrombotic risk should be taken into account, especially in the surgical setting.¹⁰¹ Treatment options of VKCFD also include recombinant activated FVII (rFVIIa), usually indicated for acquired hemophilia, congenital FVII deficiency, or Glanzmann's disease. In clinical practice, rFVIIa has been successfully used off-label for hemorrhagic conditions including warfarin overdose or severe bleeding during warfarin treatment.¹⁰²

In detail, rFVIIa was specifically administered in VKCFD during an emergency minor surgical procedure.³³ A single dose of 1.2 mg (rFVIIa ~ 20 µg/kg of body weight) allowed a successful control of hemostasis and a prompt normalization of the PT/INR. According to the different peaks of action of rFVIIa (2 hours) and VK (24 hours), their combined administration can contribute to achieve a sustained normalization of coagulation

parameters, thus representing a valid option in case of life-threatening bleeding or more complex major surgical procedures.¹⁰³ In consideration of potential life-threatening bleeding episodes, prophylaxis at diagnosis of VKCFD is strongly recommended. Usually, an oral administration of VK1 5 to 20 mg/day twice or three times per week avoids mucocutaneous bleeding. In case of poor response, VK should be administered intravenously at the dosage of 5 to 20 mg/week.⁷⁸ It is, however, known that response to replacement therapy with VK1 is widely unpredictable, depending both on the administration route and on individual sensitivity to VK.⁶¹

In vitro studies of carboxylation of differently mutated GGCX showed that some proteins have a very low or absent enzymatic activity that cannot be easily rescued with conventional doses of vitamin K.⁶¹ It is thus suggested that VKCFD1 and VKCFD2 differently respond to VK. In the meta-analysis performed by De Vilder et al, 17 patients responded well to VK treatment, 5 patients showed no or only limited response, and 3 patients did not show any response. Nine patients had no new bleeding episodes during VK prophylaxis, two patients had recurrent bleeding while on VK treatment, and the clinical outcome was not reported for six patients. In three patients, VKCFD1 was an incidental finding. In six patients, the clinical effect of VK treatment was mentioned. For five of them, a good response to VK supplementation was reported without details on the clinical outcome.⁶⁹ An in vitro study suggested that not all mutations of GGCX gene can be corrected by VK administration, in particular mutations affecting structural or catalytically important sites as the glutamate binding site (GGCX:p.(L394R), GGCX:p.(H404P)), or the propeptide binding site (GGCX:p.(R485P), GGCX:p.(G558R), GGCX:p.(T591K)).¹⁰⁴ Probably these last mutations should be managed with PCCs.¹⁰² Nevertheless, these predictive investigations are not available in clinical practice so that treatment is commonly guided by the lack of response to oral or parenteral administration of VK1.

Conclusion

Rare inherited coagulation disorders like VKCFD may challenge clinicians and laboratory specialists, in particular

for the lack of awareness of this very rare disease. To avoid misdiagnosis, the combination of comprehensive screening of residual coagulation factor activities, molecular analysis, and family segregation analysis should be performed. An accurate characterization of the disease may contribute to reach important goals in its management. However, monitoring hemostasis might be an insufficient biomarker of treatment response due to the effects of the *GGCX* mutation on several other systems. Advances in the treatment of VKCFD may not only allow to reduce bleeding complications but also offer solutions to nonbleeding manifestations of the disease. VK1 supplementation may control cardiovascular calcification and osteoporosis with an impact on the morbidity of the disease.⁸⁴ VK2 shows a longer half-life in the circulation and may be longer available for the extrahepatic tissues.¹⁰⁵ Therefore, a combined treatment with VK1 and VK2 could be considered as a valid option for managing bleeding and nonbleeding symptoms, respectively. A multidisciplinary team, including hematologists, cardiologists, and orthopedic specialists could ensure a comprehensive care for individuals with VKCFD.

Conflict of Interest
None declared.

References

- Napolitano M, Mariani G, Lapecorella M. Hereditary combined deficiency of the vitamin K-dependent clotting factors. *Orphanet J Rare Dis* 2010;5(21):21
- Darghouth D, Hallgren KW, Shtofman RL, et al. Compound heterozygosity of novel missense mutations in the gamma-glutamyl-carboxylase gene causes hereditary combined vitamin K-dependent coagulation factor deficiency. *Blood* 2006;108(06):1925–1931
- Fregin A, Rost S, Wolz W, Krebsova A, Muller CR, Oldenburg J. Homozygosity mapping of a second gene locus for hereditary combined deficiency of vitamin K-dependent clotting factors to the centromeric region of chromosome 16. *Blood* 2002;100(09):3229–3232
- Zhang B, Ginsburg D. Familial multiple coagulation factor deficiencies: new biologic insight from rare genetic bleeding disorders. *J Thromb Haemost* 2004;2(09):1564–1572
- Orphanet: Hereditary combined deficiency of vitamin K dependent clotting factors. Accessed October 29, 2023 at: https://www.orpha.net/consor/cgi-bin/OC_Exp.php?lng=EN&Expert=98434
- The *GGCX* gene homepage - Global Variome shared LOVD. Accessed February 12, 2024 at: <https://databases.lovd.nl/shared/genes/GGCX>
- Pechlaner C, Vogel W, Erhart R, Pümpel E, Kunz F. A new case of combined deficiency of vitamin K dependent coagulation factors. *Thromb Haemost* 1992;68(05):617
- Pauli RM, Lian JB, Mosher DF, Suttie JW. Association of congenital deficiency of multiple vitamin K-dependent coagulation factors and the phenotype of the warfarin embryopathy: clues to the mechanism of teratogenicity of coumarin derivatives. *Am J Hum Genet* 1987;41(04):566–583
- Vicente V, Maia R, Alberca I, Tamagnini GPT, Lopez Borrascas A. Congenital deficiency of vitamin K-dependent coagulation factors and protein C. *Thromb Haemost* 1984;51(03):343–346
- Bhattacharyya J, Dutta P, Mishra P, et al. Congenital vitamin K-dependent coagulation factor deficiency: a case report. *Blood Coagul Fibrinolysis* 2005;16(07):525–527
- Oldenburg J, von Brederlow B, Fregin A, et al. Congenital deficiency of vitamin K dependent coagulation factors in two families presents as a genetic defect of the vitamin K-epoxide-reductase-complex. *Thromb Haemost* 2000;84(06):937–941
- Spronk HM, Farah RA, Buchanan GR, Vermeer C, Soute BA. Novel mutation in the gamma-glutamyl carboxylase gene resulting in congenital combined deficiency of all vitamin K-dependent blood coagulation factors. *Blood* 2000;96(10):3650–3652
- Brenner B, Sánchez-Vega B, Wu SM, Lanir N, Stafford DW, Solera J. A missense mutation in gamma-glutamyl carboxylase gene causes combined deficiency of all vitamin K-dependent blood coagulation factors. *Blood* 1998;92(12):4554–4559
- Ghosh K, Shetty S, Mohanty D. Inherited deficiency of multiple vitamin K-dependent coagulation factors and coagulation inhibitors presenting as hemorrhagic diathesis, mental retardation, and growth retardation. *Am J Hematol* 1996;52(01):67
- Leonar CO. Vitamin K responsive bleeding disorder: a genocopy of the warfarin embryopathy. *Proceeding of the Greenwood Genetic Center* 1988;7:165–166
- Ekelund H, Lindeberg L, Wranne L. Combined deficiency of coagulation factors II, VII, IX, and X: a case of probable congenital origin. *Pediatr Hematol Oncol* 1986;3(02):187–193
- Goldsmith GH Jr, Pence RE, Ratnoff OD, Adelstein DJ, Furie B. Studies on a family with combined functional deficiencies of vitamin K-dependent coagulation factors. *J Clin Invest* 1982;69(06):1253–1260
- Puetz J, Knutsen A, Bouhasin J. Congenital deficiency of vitamin K-dependent coagulation factors associated with central nervous system anomalies. *Thromb Haemost* 2004;91(04):819–821
- Thomas A, Stirling D. Four factor deficiency. *Blood Coagul Fibrinolysis* 2003;14(Suppl 1):S55–S57
- Mickleleson KN, Whyte G. Severe deficiency of vitamin K dependent coagulation factors in an infant. *N Z Med J* 1979;90(645):291–292
- Johnson CA, Chung KS, McGrath KM, Bean PE, Roberts HR. Characterization of a variant prothrombin in a patient congenitally deficient in factors II, VII, IX and X. *Br J Haematol* 1980;44(03):461–469
- Fischer M, Zweymüller E. Kongenitaler Mangel der Faktoren II, VII und X. *Zeitschrift fuer Kinderheilkunde* 1966;95:309–323
- Arora R, Hartwig E, Kannikeswaran N. Toddler with unexplained multiple bruises. *Pediatr Emerg Care* 2014;30(09):646–648
- Al Absi HS, Abdullah MF. Congenital combined deficiency of the vitamin K-dependent clotting factors (VKCFD): a novel gamma-glutamyl carboxylase (*GGCX*) mutation. *J Pediatr Hematol Oncol* 2019;41(04):e224–e226
- Titapiwatanakun R, Rodriguez V, Middha S, Dukek BA, Pruthi RK. Novel splice site mutations in the gamma glutamyl carboxylase gene in a child with congenital combined deficiency of the vitamin K-dependent coagulation factors (VKCFD). *Pediatr Blood Cancer* 2009;53(01):92–95
- Chung KS, Bezaud A, Goldsmith JC, McMillan CW, Ménaché D, Roberts HR. Congenital deficiency of blood clotting factors II, VII, IX, and X. *Blood* 1979;53(04):776–787
- Darghouth D, Hallgren KW, Issertial O, et al. Compound heterozygosity of a W493C substitution and R704/premature stop codon within the γ -glutamyl carboxylase in combined vitamin K-dependent coagulation factor deficiency in a French family. *Blood* 2009;114(22):1302–1302
- Lunghi B, Redaelli R, Caimi TM, Corno AR, Bernardi F, Marchetti G. Novel phenotype and γ -glutamyl carboxylase mutations in combined deficiency of vitamin K-dependent coagulation factors. *Haemophilia* 2011;17(05):822–824
- Jin DY, Ingram BO, Stafford DW, Tie JK. Molecular basis of the first reported clinical case of congenital combined deficiency of coagulation factors. *Blood* 2017;130(07):948–951

- 30 Guzzardo GM, Ghosh S, Pezeshkpoor B, et al. Novel gamma-glutamyl carboxylase mutation causing hemorrhagic and non-hemorrhagic VKCFD1 phenotypes. *Blood* 2022;140(Suppl 1): 8480–8482
- 31 Rost S, Fregin A, Koch D, Compes M, Müller CR, Oldenburg J. Compound heterozygous mutations in the γ -glutamyl carboxylase gene cause combined deficiency of all vitamin K-dependent blood coagulation factors. *Br J Haematol* 2004;126(04): 546–549
- 32 McMillan CW, Roberts HR. Congenital combined deficiency of coagulation factors II, VII, IX and X. Report of a case. *N Engl J Med* 1966;274(23):1313–1315
- 33 Lapecorella M, Napolitano M, Bernardi F, et al. Effective hemostasis during minor surgery in a case of hereditary combined deficiency of vitamin K-dependent clotting factors. *Clin Appl Thromb Hemost* 2010;16(02):221–223
- 34 Boneh A, Bar-Ziv J. Hereditary deficiency of vitamin K-dependent coagulation factors with skeletal abnormalities. *Am J Med Genet* 1996;65(03):241–243
- 35 Brenner B, Tavori S, Zivelin A, et al. Hereditary deficiency of all vitamin K-dependent procoagulants and anticoagulants. *Br J Haematol* 1990;75(04):537–542
- 36 Mathonnet A, Cunat S, Allias F, et al. GGCX-related congenital combined vitamin K-dependent clotting factors deficiency-1: description of a fetus with chondrodysplasia punctata. *Am J Med Genet A* 2022;188(01):314–318
- 37 Preisler B, Pezeshkpoor B, Banchev A, et al. Familial multiple coagulation factor deficiencies (FMCDFs) in a large cohort of patients - a single-center experience in genetic diagnosis. *J Clin Med* 2021;10(02):347
- 38 Wen L, Chen J, Duan L, Li S. Vitamin K-dependent proteins involved in bone and cardiovascular health (Review). *Mol Med Rep* 2018;18(01):3–15
- 39 Kosciuszek ND, Kalta D, Singh M, Savinova OV. Vitamin K antagonists and cardiovascular calcification: a systematic review and meta-analysis. *Front Cardiovasc Med* 2022;9:938567
- 40 Li Y, Chen JP, Duan L, Li S. Effect of vitamin K2 on type 2 diabetes mellitus: a review. *Diabetes Res Clin Pract* 2018;136:39–51
- 41 Liu J, Qin L, Zheng J, et al. Research progress on the relationship between vitamins and diabetes: systematic review. *Int J Mol Sci* 2023;24(22):16371
- 42 Chen A, Li J, Shen N, Huang H, Hang Q. Vitamin K: new insights related to senescence and cancer metastasis. *Biochim Biophys Acta Rev Cancer* 2024;1879(02):189057
- 43 Xv F, Chen J, Duan L, Li S. Research progress on the anticancer effects of vitamin K2. (Review) *Oncol Lett* 2018;15(06): 8926–8934
- 44 Perrone S, De Bernardo G, Lembo C, et al. Vitamin K insufficiency and the prophylaxis strategy in term healthy infants: a multi-centre study. *Eur J Clin Invest* 2023. Doi: 10.1111/eci.14141
- 45 Vernia F, Burrelli Scotti G, Bertetti NS, et al. Low vitamin K and vitamin D dietary intake in patients with inflammatory bowel diseases. *Nutrients* 2023;15(07):1678
- 46 Sultana H, Komai M, Shirakawa H. The role of vitamin K in cholestatic liver disease. *Nutrients* 2021;13(08):2515
- 47 Wu Q, Wang L, Zhao R. Neglected vitamin K deficiency causing coagulation dysfunction in an older patient with pneumonia: a case report. *BMC Geriatr* 2022;22(01):628
- 48 Dounousi E, Liakopoulos V. Editorial for the special issue “vitamin K in chronic disease and human health”. *Nutrients* 2022;14(13):2595
- 49 Yu Z, Chen H. Piperacillin/tazobactam-induced coagulopathy in a patient through a vitamin K-dependent mechanism. *Eur J Hosp Pharm* 2021;28(04):237–238
- 50 Fotouhie A, Desai H, Parsa NA, King S. Gastrointestinal bleeding secondary to trimethoprim-sulfamethoxazole-induced vitamin K deficiency. *BMJ Case Rep* 2016;2016:bcr2016214437
- 51 Tuttolomondo A, Pignatelli P, Pola R. Editorial: Anticoagulation in cardiovascular diseases: evolving role, unmet needs, and grey areas. *Front Cardiovasc Med* 2023;10:1219033
- 52 Watzka M, Geisen C, Scheer M, et al. Bleeding and non-bleeding phenotypes in patients with GGCX gene mutations. *Thromb Res* 2014;134(04):856–865
- 53 Rydz N, James PD. The evolution and value of bleeding assessment tools. *J Thromb Haemost* 2012;10(11):2223–2229
- 54 Marchetti G, Caruso P, Lunghi B, et al. Vitamin K-induced modification of coagulation phenotype in VKORC1 homozygous deficiency. *J Thromb Haemost* 2008;6(05):797–803
- 55 Agnello L, Bellia C, Lo Coco L, et al. Vitamin K deficiency bleeding leading to the diagnosis of Crohn's disease. *Ann Clin Lab Sci* 2014;44(03):337–340
- 56 Simes DC, Viegas CSB, Araújo N, Marreiros C. Vitamin K as a diet supplement with impact in human health: current evidence in age-related diseases. *Nutrients* 2020;12(01):138
- 57 Official Journal of the European Union. Commission Directive 2008/100/EC of 28 October 2008 amending Council Directive 90/496/EEC on nutrition labelling for foodstuffs as regards recommended daily allowances, energy conversion factors and definitions. Accessed August 8, 2024 at: <https://eur-lex.europa.eu/eli/dir/2008/100/oj/eng>
- 58 Pinto E, Viegas C, Martins PV, Nascimento T, Schurgers L, Simes D. New Food Frequency Questionnaire to estimate vitamin K intake in a Mediterranean population. *Nutrients* 2023;15(13):3012
- 59 Buitenhuis HC, Soute BA, Vermeer C. Comparison of the vitamins K1, K2 and K3 as cofactors for the hepatic vitamin K-dependent carboxylase. *Biochim Biophys Acta* 1990;1034(02):170–175
- 60 Berkner KL. Vitamin K-dependent carboxylation. *Vitam Horm* 2008;78:131–156
- 61 Hao Z, Jin DY, Chen X, Schurgers LJ, Stafford DW, Tie JK. γ -Glutamyl carboxylase mutations differentially affect the biological function of vitamin K-dependent proteins. *Blood* 2021;137(04):533–543
- 62 Rishavy MA, Berkner KL. Vitamin K oxygenation, glutamate carboxylation, and processivity: defining the three critical facets of catalysis by the vitamin K-dependent carboxylase. *Adv Nutr* 2012;3(02):135–148
- 63 Chu PH, Huang TY, Williams J, Stafford DW. Purified vitamin K epoxide reductase alone is sufficient for conversion of vitamin K epoxide to vitamin K and vitamin K to vitamin KH2. *Proc Natl Acad Sci U S A* 2006;103(51):19308–19313
- 64 Liu S, Shen G, Li W. Structural and cellular basis of vitamin K antagonism. *J Thromb Haemost* 2022;20(09):1971–1983
- 65 Rost S, Fregin A, Ivaskevicius V, et al. Mutations in VKORC1 cause warfarin resistance and multiple coagulation factor deficiency type 2. *Nature* 2004;427(6974):537–541
- 66 Schulman S, Furie B. How I treat poisoning with vitamin K antagonists. *Blood* 2015;125(03):438–442
- 67 Hand I, Noble L, Abrams SA. Vitamin K and the newborn infant. *Pediatrics* 2022;149(03):e2021056036
- 68 Zhu A, Sun H, Raymond RM Jr, et al. Fatal hemorrhage in mice lacking γ -glutamyl carboxylase. *Blood* 2007;109(12):5270–5275
- 69 De Vilder EYG, Debacker J, Vanakker OM. GGCX-associated phenotypes: an overview in search of genotype-phenotype correlations. *Int J Mol Sci* 2017;18(02):240
- 70 Tie JK, Carneiro JDA, Jin DY, Martinhago CD, Vermeer C, Stafford DW. Characterization of vitamin K-dependent carboxylase mutations that cause bleeding and nonbleeding disorders. *Blood* 2016;127(15):1847–1855
- 71 Vanakker OM, Martin L, Gheduzzi D, et al. Pseudoxanthoma elasticum-like phenotype with cutis laxa and multiple coagulation factor deficiency represents a separate genetic entity. *J Invest Dermatol* 2007;127(03):581–587
- 72 Czogalla KJ, Biswas A, Rost S, Watzka M, Oldenburg J. The Arg98Trp mutation in human VKORC1 causing VKCFD2 disrupts

- a di-arginine-based ER retention motif. *Blood* 2014;124(08):1354–1362
- 73 Brenner B, Kuperman AA, Watzka M, Oldenburg J. Vitamin K-dependent coagulation factors deficiency. *Semin Thromb Hemost* 2009;35(04):439–446
 - 74 Ayyash M, Chitlur M, Oldenburg J, Shaman M. Suspected vitamin K-dependent coagulation factor deficiency in pregnancy: a case report. *Case Rep Womens Health* 2022;34:e00416
 - 75 Weston BW, Monahan PE. Familial deficiency of vitamin K-dependent clotting factors. *Haemophilia* 2008;14(06):1209–1213
 - 76 Hazim AZ, Ruan GJ, Khodadadi RB, Slusser JP, Marshall AL, Pruthi RK. A single-institution retrospective study of causes of prolonged prothrombin time and activated partial thromboplastin time in the outpatient setting. *Int J Lab Hematol* 2022;44(01):209–215
 - 77 Neutze D, Roque J. Clinical evaluation of bleeding and bruising in primary care. *Am Fam Physician* 2016;93(04):279–286
 - 78 Palla R, Peyvandi F, Shapiro AD. Congenital combined deficiency of the vitamin K-dependent clotting factors (VKCFD): a novel gamma-glutamyl carboxylase (GGCX) mutation. *Blood* 2015;125(13):2052–2061
 - 79 Mumford AD, Ackroyd S, Alikhan R, et al; BCSH Committee. Guideline for the diagnosis and management of the rare coagulation disorders: a United Kingdom Haemophilia Centre Doctors' Organization guideline on behalf of the British Committee for Standards in Haematology. *Br J Haematol* 2014;167(03):304–326
 - 80 Tcherniantchouk O, Laposata M, Marques MB. The isolated prolonged PTT. *Am J Hematol* 2013;88(01):82–85
 - 81 Mathews N, Tasneem S, Hayward CPM. Rare inherited coagulation and fibrinolytic defects that challenge diagnostic laboratories. *Int J Lab Hematol* 2023;45(Suppl 2):30–43
 - 82 Suttie JW. Vitamin K and human nutrition. *J Am Diet Assoc* 1992;92(05):585–590
 - 83 Shearer M. The assessment of human vitamin K status from tissue measurements. In: Suttie JW, ed. *Current Advances in Vitamin K Research*. New York, NY:1985
 - 84 Fusaro M, Gallieni M, Rizzo MA, et al. Vitamin K plasma levels determination in human health. *Clin Chem Lab Med* 2017;55(06):789–799
 - 85 Card DJ, Gorska R, Harrington DJ. Laboratory assessment of vitamin K status. *J Clin Pathol* 2020;73(02):70–75
 - 86 Sokoll LJ, Sadowski JA. Comparison of biochemical indexes for assessing vitamin K nutritional status in a healthy adult population. *Am J Clin Nutr* 1996;63(04):566–573
 - 87 Shaw MA, Liu A. Take the shot: a review of vitamin K deficiency. *Pediatr Ann* 2023;52(02):e42–e45
 - 88 Loyal J, Shapiro ED. Refusal of intramuscular vitamin K by parents of newborns: a review. *Hosp Pediatr* 2020;10(03):286–294
 - 89 Shao X, Ren Y, Xie N, et al. Effect of cefoperazone/sulbactam on blood coagulation function in infected emergency department patients and the necessity of vitamin K1 (VK1) preventive intervention: a single-center, retrospective analysis. *Med Sci Monit* 2023;29:e939203
 - 90 Tassew SF, Haile BA, Amara Birlie T. Outcome of rodenticide poisoning and its associated factors among adult patients admitted with rodenticide poisoning at the emergency unit of Debre Tabor Comprehensive Specialized Hospital, Debre Tabor, North Central Ethiopia. *Open Access Emerg Med* 2023;15:189–197
 - 91 van Breemen RB, Hafner JW, Nosal DG, Feinstein DL, Rubinstein I. Unmet clinical laboratory need in patients hospitalized for acute poisoning from long-acting anticoagulant rodenticides. *Toxicol Commun* 2021;5(01):93–96
 - 92 Eden RE, Daley SF, Coviello JM. Vitamin K Deficiency. In: *StatPearls*. StatPearls Publishing; 2024 Accessed February 14, 2024 at: <http://www.ncbi.nlm.nih.gov/books/NBK536983/>
 - 93 Li T, Chang CY, Jin DY, Lin PJ, Khvorova A, Stafford DW. Identification of the gene for vitamin K epoxide reductase. *Nature* 2004;427(6974):541–544
 - 94 Wu SM, Stafford DW, Frazier LD, et al. Genomic sequence and transcription start site for the human gamma-glutamyl carboxylase. *Blood* 1997;89(11):4058–4062
 - 95 Wu SM, Morris DP, Stafford DW. Identification and purification to near homogeneity of the vitamin K-dependent carboxylase. *Proc Natl Acad Sci U S A* 1991;88(06):2236–2240
 - 96 Wu SM, Cheung WF, Frazier D, Stafford DW. Cloning and expression of the cDNA for human gamma-glutamyl carboxylase. *Science* 1991;254(5038):1634–1636
 - 97 Hosseini MS, Shams M, Dorgalaleh A, Mansouritorghabeh H. Multiple Coagulation Factor Deficiency. In: Dorgalaleh A, ed. *Congenital Bleeding Disorders*. Springer International Publishing; 2018:219–238. Doi: 10.1007/978-3-319-76723-9_9
 - 98 Chen HF, Wu TQ, Jin LJ, et al. Treatment of vitamin K-dependent coagulation factor deficiency and subarachnoid hemorrhage. *World J Emerg Med* 2011;2(01):73–76
 - 99 Zeng L, Choonara I, Zhang L, Li Y, Shi J. Effectiveness of prothrombin complex concentrate (PCC) in neonates and infants with bleeding or risk of bleeding: a systematic review and meta-analysis. *Eur J Pediatr* 2017;176(05):581–589
 - 100 World Health Organization. Global status report on blood safety and availability 2021. Accessed February 14, 2024 at: <https://www.who.int/publications-detail-redirect/9789240051683>
 - 101 Lubetsky A, Hoffman R, Zimlichman R, et al. Efficacy and safety of a prothrombin complex concentrate (Octaplex) for rapid reversal of oral anticoagulation. *Thromb Res* 2004;113(06):371–378
 - 102 Leissinger CA, Blatt PM, Hoots WK, Ewenstein B. Role of prothrombin complex concentrates in reversing warfarin anticoagulation: a review of the literature. *Am J Hematol* 2008;83(02):137–143
 - 103 Ng HJ, Lee LH. Recombinant activated clotting factor VII (rFVIIa) in the treatment of surgical and spontaneous bleeding episodes in hemophilic patients. *Vasc Health Risk Manag* 2006;2(04):433–440
 - 104 Ghosh S, Kraus K, Biswas A, et al. GGCX mutations show different responses to vitamin K thereby determining the severity of the hemorrhagic phenotype in VKCFD1 patients. *J Thromb Haemost* 2021;19(06):1412–1424
 - 105 Halder M, Petsophonsakul P, Akbulut AC, et al. Vitamin K: double bonds beyond coagulation insights into differences between vitamin K1 and K2 in health and disease. *Int J Mol Sci* 2019;20(04):896